

Real-Time Remote Cancellation of Multi-Tones In An Extended Acoustic Cavity Using Directional Ultrasonic Loudspeaker

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Abstract— This paper presents an adaptive active noise control (ANC) method to remotely cancel multi-tone noise signals in a room using ultrasonic ‘AudioSpotLight’ (ASL) as the noise canceling loudspeaker. The ASL is a directional parametric array loudspeaker whose 60 kHz carrier frequency is modulated by the anti-noise signal generated by adaptive ANC algorithm. The proposed method uses no secondary path estimate and is capable of canceling the tonal noise signals at distant target points, in real time. Using adaptive Leaky-NLMS algorithm, noise attenuation levels (NAL) of up to 15.71 dB are obtained in the presence of diffused low frequency background noise. Performance analysis of the proposed system is presented and the results of a simple real-time implementation are also provided to support the proposed ANC method.

Keywords— Adaptive NLMS algorithm, Active Noise control, Remote noise cancellation, Ultrasonic loudspeaker.

I. INTRODUCTION

Active noise control (ANC) has been an important topic of research for over three decades and has witnessed advances in terms of architectures, algorithms and applications. ANC systems are a very attractive and effective way to suppress unwanted noise present in several industrial environments and consumer applications. ANC is based on the principle of ‘acoustic’ superposition of a noise signal and a synthetic ‘anti-noise’ signal of same amplitude and exactly opposite phase, using appropriate adaptive algorithms, to achieve maximum attenuation of noise in particular zone of interest. However, difficulties have persisted in real-time implementation of many ANC systems due to acoustic path delay and interactions, as well as computational burden imposed on the processors. Also, most ANC systems deployed over the span of past several years focus on controlling noise in a rather small confined acoustic cavity, like ducts, tubes, cabinets, ear cups, avoiding or reducing the negative effects of long delay, dispersion of acoustic signal, interference of multiple signal paths, and background noise on the convergence, stability and performance of adaptive ANC algorithms. These aforementioned constraints combined finally limit the practical deployment of ANC systems in larger acoustic environments like rooms, hallways, concert halls, etc. This paper presents a simple yet robust method to design and deploy a real-time ANC system in an extended acoustic

cavity. The ANC algorithm driving the ANC system has to be simple, computationally fast, yet robust against changes in the environment. Hence, we use the most popular Normalized Least Means Square (NLMS) adaptive algorithm used for ANC applications and modify one of the parameters -‘leaky factor’ for our application to introduce further stability in changing acoustic environments. The results obtained from the above algorithm are presented and performance of the ANC system is analyzed. ANC is an attractive means to achieve large noise reduction for low to relatively high frequency, for both and slow varying semi-periodic and periodic noise signals [1][2][3][4]. We have used pure periodic tones from 1 to 9 kHz as noise signals in our experiments, for the following reasons: (1) Periodic signals are predictable and theoretically can be cancelled very well, (2) Exact secondary path estimation can be ignored, which reduces computational burden. (3) Due to non-linearity of human ear and psychoacoustics, the human ear is more sensitive to frequencies in this band (1- 9 kHz) Hence controlling these frequencies is more promising to achieve a better perceivable noise attenuation. (4) In many industrial applications, the generated noise has periodic/semi-periodic nature (from cooling fans, compressors, turbines, etc.) (5) They are easier to handle. The attenuation results for single tones are further extended to multi-tones and presented in this paper. This paper also sheds some light on the setup selection for ANC hardware viz. the relative placement of noise source loudspeaker, noise canceling (secondary source) loudspeaker, error microphone and reference microphone, for achieving maximum attenuation.

Next novel aspect of this paper is the successful use of directional ultrasonic ASL loudspeaker [11] as the noise canceling loudspeaker for our ANC application. ASL loudspeaker (by Holosonics) is a directional parametric array loudspeaker. ASL works as follows. The 60 kHz carrier frequency of ASL is modulated by a desired audio signal, amplified and propagated through directional pattern of the ASL. The modulated signal is then automatically demodulated, by reflection and non-linearity properties, at the target point and the modulating audio signal is heard at the object or target point within the ASL directional pattern. As our experiments have shown, besides automatic signal demodulation, and unlike a regular or traditional

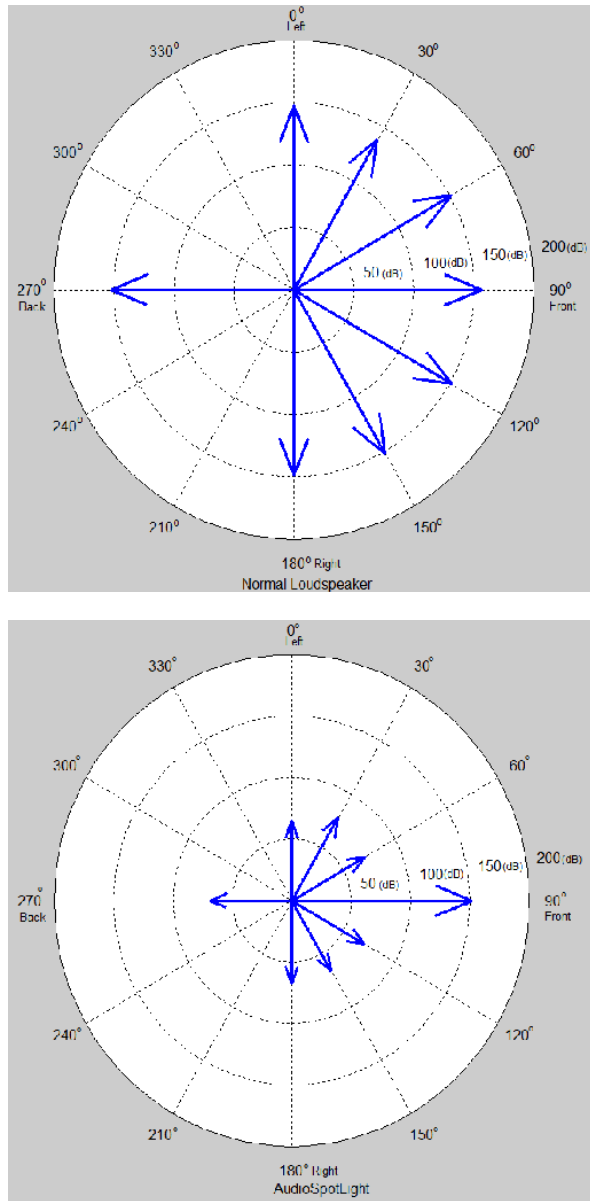


Fig. 1. Comparison of variation in intensity of a 2 kHz signal around a 106 cm circle between (a) Normal Loudspeaker and (b) directional ASL speaker. Length of the arrow represents the intensity of sound pressure level.

Loudspeaker, the power and intensity of the original modulating audio signal degrades very little when appeared at considerable distances (e.g. up to 73 meters). The spatial characteristics of the directional ASL is shown in Fig 1. We use these properties to generate anti-noise signal as the modulating signal for ASL to cancel periodic noise signals at a distant location by adaptive ANC algorithms. This approach is taken since in many ANC applications, it is difficult to place the cancelling loud-speaker very close to the target point in cancellation zone thus, limiting the interaction of generated

anti-noise with the noise near the error microphone and degrading the overall performance of the ANC system. The adaptive ANC proposed in this paper uses the ASLs as the cancelling loudspeakers, which can direct anti-noise signal to remotely cancel noise signal over long line-of-sight distances. The use of ASL as a cancelling loudspeaker provides a more localized quiet zone, without increasing the adjacent sound pressure levels [5], unlike the traditional loudspeakers used in the previous ANCs. While some authors [6] have argued the inability of ASL to be used as for ANC applications, the results presented here show possibility of using ultrasonic loudspeakers to remotely cancel or control undesired signals at some distant points. A brief outline of the modulation process employed by ASL is provided in this paper which would help to better understand ASL operation, advantages, and disadvantages. In our experiments, single-channel adaptive leaky-NLMS-based feedforward ANC is implemented on a PC to cancel narrowband (periodic) noise signals at a distant target point using ultrasonic ASL loudspeakers in an extended acoustic cavity.

II. RELATION TO PRIOR WORK

F. Joseph Pompei, in 2002, introduced the ASL loudspeakers using ultrasound as an audible source in his Ph.D. thesis [7]. Since then, numerous researchers have investigated the feasibility of ASL as cancelling loudspeaker for ANC system. In 2005, Brooks et al. [6] and Tan and Tanaka [8] independently studied the ASL loudspeaker and concluded that, owing to a high noise floor at low frequencies, it was only applicable for higher frequencies (above 1 kHz). In 2009, Tanaka and Tanaka [9] came up with a steerable parametric array and suppressed a 1.5 kHz sinusoid by 13.7dB SPL at a separation of 0.6m from cancelling loudspeaker. Ye et al. [10] used a 10cm diameter parametric array in a straight line arrangement to suppress a 2 kHz sinusoid by 6db SPL, in an anechoic chamber, in 2012. Our paper presents a leaky NLMS based-ANC, using ASL loudspeakers, to cancel tonal noise in an extended acoustic cavity (e.g. a room) in presence of low frequency background noise, from a distance of 106cm (~42 inches), requiring no secondary path estimate. NAL (Noise Attenuation Level) values up to 15.71dB was achieved using this approach.

III. METHODS

An important assumption in this paper is the selection of noise signals to be periodic. Periodic sequences, such as sinusoids, are predictable and the knowledge of their one period is enough to analyze the entire sequence at any time. This in turn enables overcoming time delays from acoustic wave propagation, algorithms, and the entire ANC system. Furthermore, assuming linear phase propagation in the acoustic medium (free air), the frequency (and hence wavelength) of a sinusoid can be related to measureable distance and delay. This helps us to estimate the points in space, where the sinusoidal signal pressure will be minimum or maximum. For example, assuming the speed of sound of 340m/sec in air, a 2 kHz sinusoid (with zero phase) has a wavelength of ~17 cm. Hence, there is a point of minimum

sound pressure level at every 17 cm, corresponding to a phase shift 2π radians from the source. Using this simple property, if we can simultaneously generate a delayed version of the sinusoid, delayed by one half of the wavelength (phase shift by π radians), theoretically these sinusoids will destructively superimpose in the acoustic domain and there will be minimum sound pressure level in the region of their interaction. This is the idea behind using pure periodic signals as test signals for this analysis which can then be expanded to cancel narrow band signals. While the concept of attenuation using phase delay is lucid, it is not very practical and robust for real-time cancellation of variable noise signals. So we need an algorithm that is adaptive to changes in the noise amplitude and yet stable against unwanted variations noise characteristics. So we adopt the leaky-NLMS (Normalized Least Mean Square) adaptive algorithm for our use. If $\mathbf{x}(n)$ is the input noise vector of length Q and $y(n)$ is the anti-noise for $x(n)$, the adaptive leaky NLMS algorithm is represented by the following equations:

$$y(n) = \sum_{l=0}^{L-1} \mathbf{w}_l(n) \mathbf{x}(n-l) \quad (1)$$

$$\mathbf{w}(n+1) = v \mathbf{w}(n) + \frac{\mu \mathbf{x}(n)e(n)}{\|\mathbf{x}(n)\|^2} \quad (2)$$

where, for time samples $n \geq 0$, $L \leq Q$, step size $0 \leq \mu < 1$, $\mathbf{w}(n)$ vector coefficients of the adaptive filter $\mathbf{W}(z)$ with elements $\mathbf{w}(n)$, error signal $e(n)$ at the error microphone, and the leaky factor $0 \leq v \leq 1$. In (2), the leaky factor v is a modification to the cost function that constraints the filter coefficients and improves the algorithm's stability [1], which is justified by the experiment results provided in Table 2.

In this paper, the measure NAL (dB) is calculated as follows:

$$\text{NAL(dB)} = 10 * \log_{10} \left[\frac{\|\mathbf{x}(n)\|^2}{\|\mathbf{e}(n)\|^2} \right] \quad (3)$$

In the above expression, $\|\mathbf{x}(n)\|^2$ represents the power of the noise signal (before attenuation) and $\|\mathbf{e}(n)\|^2$ represents the power of the error signal after attenuation. Alternatively, the dB SPL measure could also be used. Inter conversion from NAL (dB) to SPL (dB) is also provided in [4].

As discussed in [5][6][7], the ASL (also known as parametric array loudspeaker), that we use as canceling loudspeaker, can generate a highly directional sound beam, with carrier frequency $\omega_c \approx 60\text{kHz}$ (in Radian per Sec: $\omega_c = 2\pi 60000$), and modulated by desired audible signal. It is known that a sound beam becomes highly directional when the source size is comparable to the wavelength of the sound. This suggest (and is widely available in literature) (see references in [7]) that at higher frequencies, sound becomes highly directional. We begin by reproducing the Khokhlov-Zabolotskaya-Kuznetsov (KZK) equation that accounts for the combined effects of non-linearity, diffraction and absorption in directional sound beams, and is described by:

$$\frac{\partial^2 p}{\partial z \partial r} = \frac{c_0}{2} \nabla_r^2 p + \frac{\delta}{2c_0^3} \frac{\partial^3 p}{\partial r^3} + \frac{\beta}{2p_0 c_0^3} \frac{\partial^2 p^2}{\partial r^2} \quad (4)$$

where p is the audible sound pressure, c_0 is the small signal sound speed, δ is the sound diffusivity, β is the non-linearity coefficient and p_0 is the ambient density. Notice that the first two terms: $\frac{c_0}{2} \nabla_r^2 p + \frac{\delta}{2c_0^3} \frac{\partial^3 p}{\partial r^3}$ (say P_1) are linear in p and their solution is related to low-distortion audio reproduction, while the last term $\frac{\beta}{2p_0 c_0^3} \frac{\partial^2 p^2}{\partial r^2}$ (say P_2) is non-linear in p and its solution relates to predictable audible beam geometry. One set of straightforward approximate solution for p is given by (as suggested in [7]):

$$p_1 = [p_0 E(t) \sin \omega_c t] e^{-ax} \quad , a > r \quad (5)$$

$$p_2 = \frac{\beta p_0^2 a^2}{16 \rho_0 \alpha c_0^4 x} \frac{\partial^2 E^2(\tau)}{\partial \tau^2} \quad (6)$$

where a is the dimension of the transducer, α is the ultrasonic absorption ($\propto \omega_c^2$), ω_c is the ultrasonic carrier frequency and $E(t)$ is the modulation envelope of the amplitude modulated (AM) signal. A lot of valuable information can be derived from equation (5) and (6):

- (i) p_1 represents a damped AM signal.
- (ii) $p_2 \propto p_0^2$, which implies audible pressure levels are non-linearly related to the ultrasonic pressure levels.
- (iii) $p_2 \propto a^2$, which implies audible pressure levels are directly related to the area of the transducer.
- (iv) $p_2 \propto \frac{\partial^2 E^2(\tau)}{\partial \tau^2}$, which implies the audible pressure levels have a +12db/octave spectral slope. This means lower frequencies need more power to be reproduced. Also the audible pressure levels have a square relationship with the modulation envelope, $E(t)$.

The high directionality of modulated ultrasonic sound beam can be attributed to the careful selection of carrier frequency (for ASL, $\omega_c \approx 60\text{ kHz}$) which has minimum ultrasonic absorption, α with distance.

If $g(t)$ is the modulating signal (with modulating frequency, ω_0), a reasonable assumption for $E(t)$ can be of the following form [7]:

$$E(t) = [L(t) + m \iint g(\tau) d\tau d\tau]^{0.5} \quad (7)$$

Here, m is the modulation index ($m < 1$) and $L(t)$ is a temporally asymmetric offset function. The exponent 0.5 in (7) introduces harmonics in the modulation process. In this paper, we propose the use of anti-noise signal (output of leaky NLMS filter, $y(n)$) as the modulating signal, $g(t)$.

The demodulation process is fairly simple. Due to non-linear propagation through air and due to reflection from nearby objects, the AM signal gets 'self-demodulated' to the modulating frequency, and falls into the human audible range.

IV. EXPERIMENTS AND RESULTS

In this section, we present the hardware used and the experiments which led us to build a robust yet simple ANC system for remotely cancelling tonal noise signals. In all our experiments, No secondary path estimate, as well-known in ANC systems, was used. The reference noise signal was generated through the Polk Audio loudspeaker

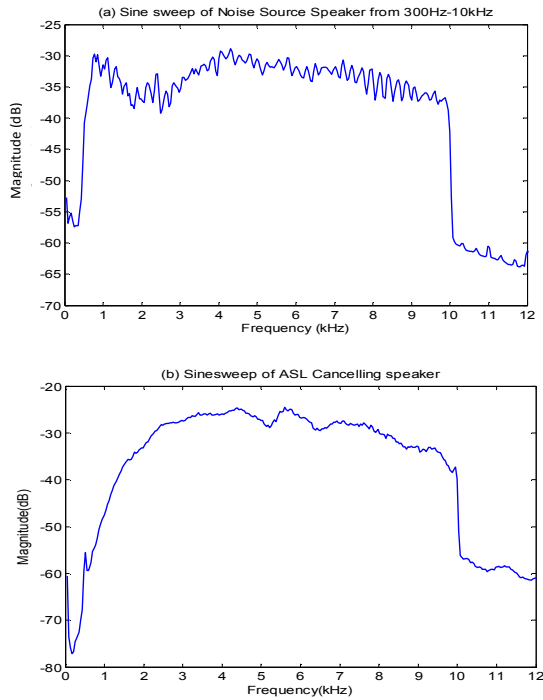


Fig. 2. Lumped Magnitude spectra of (a) Noise source loudspeaker, (b) Cancelling loudspeaker, including Band pass filters, Microphones, ADC and DAC.

RC65i and the AudioSpotLight ASL 24 ultrasonic loudspeaker operating at carrier frequency of 60 kHz was used as the noise canceling loudspeaker. Pre-polarized Diffuse-field $\frac{1}{2}$ " B & K type 4942C microphones were used as the reference and error microphones. The leaky NLMS algorithm was coded in MATLAB 2013a and modelled using Simulink on a Dell Latitude E6530 laptop (Windows 7 platform). The dimensions of the room were (length x width x height) 5.79m x 4.57 m 2.74 m (Fig.4) and variance of background noise was 0.0091. Signal-to-noise ratio (SNR $\sim$ 20.4dB) was kept reasonably high. The lumped magnitude spectra of the noise source loudspeaker, cancelling ASL loudspeaker (obtained for a Sine sweep from 300 to 10kHz) are shown in Fig. 2. The effects of these spectra on experimental results will be discussed in subsequent sections.

A. Selection of Best Setup for ANC system by manual phase delay method

In this first experiment, the goal is to compare three setups shown in Fig. 3, for the relative placement of reference noise loudspeaker, canceling loudspeaker and the error microphone so as to achieve maximum noise attenuation. As described in Section 3, a delay expressed as a phase shift was obtained for every single tone. The phase-shifted version of noise tone was then used as anti-noise (the modulating signal for the ultrasonic loudspeaker) and the NAL was measured for every noise tone as shown in Table 1. Next, we used sums of sinusoids as reference noise, and the anti-noise is formed by delaying each

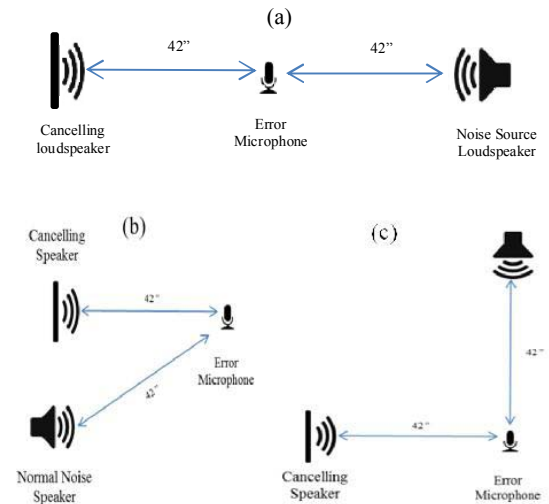


Fig. 3. Schematic diagram showing three setups compared on basis of NAL. (a) Straight line, facing each other (b) Straight line, facing same direction (c) Orthogonal setup. A separation of 42" between source and error microphone is chosen arbitrarily.

constituent single sinusoid by its known phase shift value. Table 1 shows that the Setup (c) performs better than the rest two in terms of NAL. This is attributed to the interaction between the noise sound waveform propagation and the anti-noise waveform propagation at the error microphone. Henceforth, we adopt the third setup (Fig. 3. (c)) in rest of our experiments.

The values of NAL in Table 1 can also be treated as the ground truth for further comparison, assuming the reference noise is time-invariant. Most importantly, these experimental results, although straightforward, strongly suggest that it is possible to use ASL as a secondary source loudspeaker to cancel single tone as well as multi-tone signals. This encouraged us to develop and employ the leaky NLMS to adaptively cancel the single and multiple tones.

TABLE I. COMPARISON OF NAL VALUES FOR THREE SETUPS.

Frequency (Hz)	Phase Shift	NAL (dB)		
		Setup (a)	Setup (b)	Setup (c)
2000	1.2π	11.22	12.57	14.08
3000	1.4π	12.21	16.04	18.42
6000	1.4π	10.97	10.85	15.51
8000	1.6π	8.56	13.77	16.42
2k+3k+4k+5k+6k+7k		11.18	10.11	12.00
2k+3k+4k+5k+7k+9k		10.01	9.13	12.57

B. Real-time implementation of NLMS based ANC for suppressing Tonal noise.

Now, we focus our attention to the real time scenario, where the leaky-NLMS algorithm is deployed to attenuate the tonal noise in the aforementioned extended cavity without prior

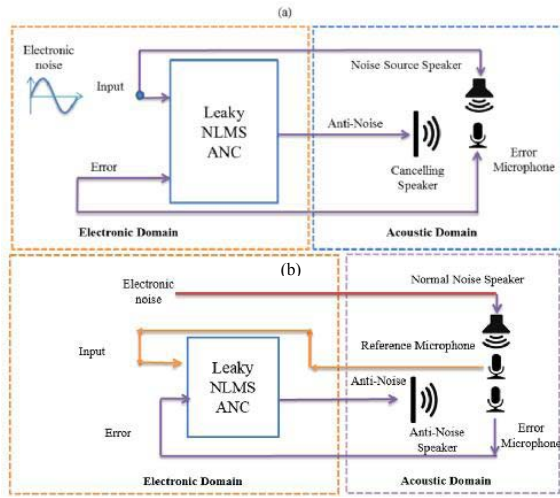


Fig. 4. Schematic diagram (a) Model 1 and (b) Model 2.

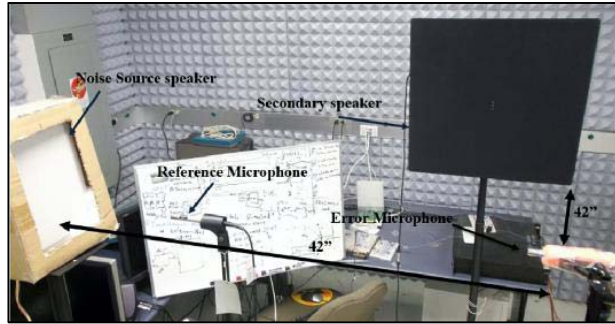


Fig. 5. Experimental setup for Model 2. Due to space constraints, the separation between error microphone and the cancelling (secondary) loudspeaker was restricted to 1.06m (~42").

knowledge of the phase/delay of reference periodic noise signals. Tonal noise is played through the noise source loudspeaker and the anti-noise generated by the Leaky NLMS algorithm is used as the modulating signal for the ASL loudspeaker. Error signal is picked up by the microphone and goes through a band pass filter (to remove most of the low frequency noise). It is then sampled by an analog to digital converter (ADC) at 44.1 kHz and interfaced to the Simulink model. The Simulink model down-samples the input data to 20 kHz before feeding it sample-by-sample to the adaptive Leaky NLMS algorithm running on the PC. The down sampling frequency is selected in reference to the highest frequency of interest (9 kHz), based on the Nyquist theorem. The experiment is divided into two models: In Model 1 shown in Fig. 4 (a), tonal noise is provided 'electronically', i.e. not measured by a reference microphone, to the adaptive NLMS algorithm and the error signal is read 'acoustically' using the error microphone and given to the algorithm. Since, we provide the 'electronic' noise to the NLMS algorithm instead of the 'acoustic' noise for modelling the filter weights, a lower order filter is sufficient to produce good NAL in this model

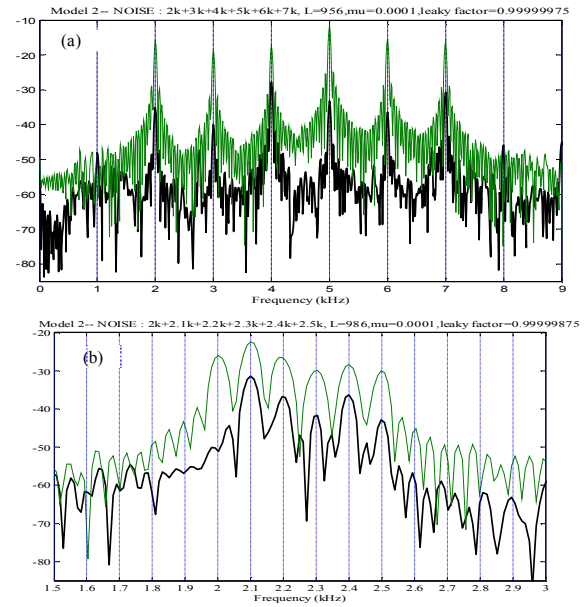


Fig. 6. Magnitude spectrum of Sum of sinusoids (a) placed 1 kHz apart (b) placed 100 Hz apart before attenuation (light line) and after attenuation (dark line) for Model 2.

In Model 2 (shown in Fig. 4(b)), both noise and error signals are read 'acoustically' using reference microphone and error microphone, respectively. Model 2 is a practical case of Model 1, because the reference signal read from another microphone, is already corrupted by the background noise present in the room. By increasing the order of adaptive filter, the leaky-NLMS algorithm models the 'acoustic' noise quite well and the ANC system results in good cancellation of reference noise signals. Fig. 5 shows the actual experimental set up for Model 2 showing the noise source loudspeaker, cancelling loudspeaker, the reference microphone and error microphone. The results from these experiments are presented and compared for Model 1 and Model 2 in Table 2. Table 2 also shows the sensitivity of leaky factor on the convergence of the NLMS in the presence of background noise. Fig. 6 (a) and 6(b) presents the magnitude spectra of signals before attenuation and after attenuation for Model 2. Within limits of acceptable harmonic distortion introduced by the modulation process of ASL (see Fig. 2 (b)), the lowest usable frequency is 1 kHz and the highest usable amplitude depends on the tonal noise amplitude. The latter is due to the non-linear gain limit of the ASL modulating amplifier. So care was taken to perform all the experiments in a very restricted frequency-amplitude range so as to minimize harmonic distortion and still have maximum attenuation. Although all the generated reference sinusoids were of equal amplitudes, the amplitude differences in the magnitude spectra of attenuated noise signals are due to the 'non-flat' magnitude spectra of the loudspeakers as shown in Fig. 2(a), (b) and due to the background noise in the room.

TABLE II. COMPARISON OF NAL VALUES FOR MODEL 1 AND MODEL 2 WITH $\mu=0.0001$.

Frequency (Hz)	Model 1			Model 2		
	L	ν	NAL (dB)	L	ν	NAL (dB)
2000	32	0.999995	12.76	48	0.999997	15.04
3000	32	0.999995	12.21	48	0.999995	13.28
6000	32	0.999995	10.05	64	0.999945	13.44
8000	64	0.999997	11.92	128	0.999957	15.71
2k+3k+4k+5k+6k+7k	850	0.9999985	16.59	956	0.99999975	9.00
2k+3k+4k+5k+7k+9k	874	0.9999983	14.81	986	0.99999977	9.62
2k+2.1k+2.2k+2.3k+2.4k+2.5k	850	0.9999987	13.05	986	0.99999875	11.79

L=Filter order, ν =Leaky factor, μ =Step size.

V. CONCLUSIONS

Present day ANC systems target controlling noise in confined acoustic environments, by placing cancelling loudspeakers close to the zone of cancellation. In this paper, an adaptive real-time remote noise control technique using ultrasonic ASL loudspeakers was proposed for a room. A leaky NLMS based ANC algorithm was employed as the control algorithm. No secondary path estimate was used. Under limitations of ASL hardware, experimental evaluations of the proposed method show promising noise attenuation levels ranging from 9 to 15.71dB were obtained for tonal noise signals from 1 to 9 kHz. The algorithm development and testing was performed using a laptop PC (with necessary support hardware), implying computationally fast processing and cost-efficient industrial solution.

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